

Project Design Phase-II  
Solution Requirements (Functional & Non-functional)

|               |                                                           |
|---------------|-----------------------------------------------------------|
| Date          | 27 February 2026                                          |
| Team ID       | LTVIP2026TMIDS84546                                       |
| Project Name  | Explore with AI: Custom Itineraries for Your Next Journey |
| Maximum Marks | 4 Marks                                                   |

Functional Requirements:

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic)    | Sub Requirement (Story / Sub-Task)                                       |
|--------|----------------------------------|--------------------------------------------------------------------------|
| FR-1   | User Input Interface             | Enter travel destination                                                 |
|        |                                  | Enter trip duration (days/nights)                                        |
|        |                                  | Select travel interests (sightseeing, adventure, food, relaxation, etc.) |
| FR-2   | AI Itinerary Generation          | Generate a day-wise personalized itinerary based on user inputs          |
|        |                                  | Include attractions, dining, and travel tips                             |
| FR-3   | Itinerary Review & Customization | Review the generated itinerary                                           |
|        |                                  | Modify or adjust activities, timing, or remove items                     |
| FR-4   | Export & Sharing                 | Export itinerary as PDF or copy to clipboard                             |
| FR-5   | Error Handling                   | Validate inputs and provide user-friendly error messages                 |
| FR-6   | Optional Recommendations         | AI suggests hidden gems or additional attractions based on preferences   |

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

| NFR No. | Non-Functional Requirement | Description |
|---------|----------------------------|-------------|
|---------|----------------------------|-------------|

|       |               |                                                                                                                     |
|-------|---------------|---------------------------------------------------------------------------------------------------------------------|
| NFR-1 | Usability     | The app must offer a simple, intuitive interface for users of all experience levels                                 |
| NFR-2 | Security      | API calls to Gemini Pro should be secure; inputs/outputs should be safely handled                                   |
| NFR-3 | Reliability   | The system must generate itineraries consistently without crashes                                                   |
| NFR-4 | Performance   | AI-generated itineraries should be returned in under 5 seconds (depending on destination and complexity)            |
| NFR-5 | Availability  | The platform should maintain high uptime ( $\geq 99.5\%$ )                                                          |
| NFR-6 | Scalability   | System should handle multiple users generating itineraries simultaneously                                           |
| NFR-7 | Extensibility | System should allow future integration with features like multi-destination trips, chatbots, or travel booking APIs |